

```
mysql> use cg_mysql;
Database changed
```

```
mysql> CREATE TABLE Employees (
->     emp_id INT PRIMARY KEY,
->     name VARCHAR(100),
->     department_id INT,
->     salary DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql>
mysql> CREATE TABLE Departments (
->     department_id INT PRIMARY KEY,
->     department_name VARCHAR(100)
-> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql>
mysql> CREATE TABLE Projects (
->     project_id INT PRIMARY KEY,
->     project_name VARCHAR(100),
->     emp_id INT
-> );
Query OK, 0 rows affected (0.04 sec)
```

```
mysql>
mysql> INSERT INTO Departments (department_id, department_name) VALUES
-> (1, 'Human Resources'),
-> (2, 'Engineering'),
-> (3, 'Marketing'),
-> (4, 'Finance'),
-> (5, 'Operations');
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql>
mysql>
mysql> INSERT INTO Employees (emp_id, name, department_id, salary) VALUES
-> (101, 'Arjun', 1, 50000),
-> (102, 'Bhavana', 2, 65000),
-> (103, 'Charan', 2, 72000),
-> (104, 'Divya', 3, 45000),
-> (105, 'Eswar', NULL, 40000),
-> (106, 'Farhan', 4, 55000),
-> (107, 'Gowri', 2, 62000),
-> (108, 'Harish', 5, 48000),
-> (109, 'Ishita', NULL, 53000),
-> (110, 'Jagan', 3, 60000);
```

Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql>

mysql>

```
mysql> INSERT INTO Projects (project_id, project_name, emp_id) VALUES
-> (201, 'Website Redesign', 102),
-> (202, 'Mobile Banking App', 103),
-> (203, 'Marketing Campaign', 104),
-> (204, 'Payroll System', 106),
-> (205, 'Backend Optimization', 107),
-> (206, 'Cloud Migration', 102),
-> (207, 'Customer Analytics', 110),
-> (208, 'Inventory System', NULL);
```

Query OK, 8 rows affected (0.01 sec)
Records: 8 Duplicates: 0 Warnings: 0

mysql> select * from employees;

emp_id	name	department_id	salary
101	Arjun	1	50000.00
102	Bhavana	2	65000.00
103	Charan	2	72000.00
104	Divya	3	45000.00
105	Eswar	NULL	40000.00
106	Farhan	4	55000.00
107	Gowri	2	62000.00
108	Harish	5	48000.00
109	Ishita	NULL	53000.00
110	Jagan	3	60000.00

10 rows in set (0.02 sec)

mysql> select * from departments;

department_id	department_name
1	Human Resources
2	Engineering
3	Marketing
4	Finance
5	Operations

5 rows in set (0.00 sec)

mysql> select * from projects;

project_id	project_name	emp_id
------------	--------------	--------

201	Website Redesign	102
202	Mobile Banking App	103
203	Marketing Campaign	104
204	Payroll System	106
205	Backend Optimization	107
206	Cloud Migration	102
207	Customer Analytics	110
208	Inventory System	NULL

8 rows in set (0.00 sec)

mysql> -- get all employees with their departments.

mysql> select e.emp_id, e.name, d.department_name from employees e inner join departments d on e.department_id = d.department_id;

emp_id	name	department_name
101	Arjun	Human Resources
102	Bhavana	Engineering
103	Charan	Engineering
104	Divya	Marketing
106	Farhan	Finance
107	Gowri	Engineering
108	Harish	Operations
110	Jagan	Marketing

8 rows in set (0.00 sec)

mysql> -- get all employees who was assigned to a project

mysql> select e.emp_id, e.name, p.project_name from employees e inner join projects p on e.emp_id = p.emp_id;

emp_id	name	project_name
102	Bhavana	Website Redesign
103	Charan	Mobile Banking App
104	Divya	Marketing Campaign
106	Farhan	Payroll System
107	Gowri	Backend Optimization
102	Bhavana	Cloud Migration
110	Jagan	Customer Analytics

7 rows in set (0.00 sec)

mysql> select e.emp_id, e.name from employees e inner join projects p on e.emp_id = p.emp_id;

emp_id	name
102	Bhavana

103	Charan
104	Divya
106	Farhan
107	Gowri
102	Bhavana
110	Jagan

+-----+-----+

7 rows in set (0.00 sec)

```
mysql> select distinct e.emp_id, e.name from employees e inner join projects p on
e.emp_id = p.emp_id;
```

+-----+-----+

emp_id	name
102	Bhavana
103	Charan
104	Divya
106	Farhan
107	Gowri
110	Jagan

+-----+-----+

6 rows in set (0.00 sec)

```
mysql> select e.emp_id, e.name, d.department_name from employees e left join
departments d on e.department_id = d.department_id;
```

+-----+-----+-----+

emp_id	name	department_name
101	Arjun	Human Resources
102	Bhavana	Engineering
103	Charan	Engineering
104	Divya	Marketing
105	Eswar	NULL
106	Farhan	Finance
107	Gowri	Engineering
108	Harish	Operations
109	Ishita	NULL
110	Jagan	Marketing

+-----+-----+-----+

10 rows in set (0.00 sec)

```
mysql> select emp_id, name from employees where department_id is null;
```

+-----+-----+

emp_id	name
105	Eswar
109	Ishita

+-----+-----+

2 rows in set (0.00 sec)

```
mysql> select e.emp_id, e.name, p.project_name from employees e left join projects
p on e.emp_id = p.emp_id where project_name is null;
```

emp_id	name	project_name
101	Arjun	NULL
105	Eswar	NULL
108	Harish	NULL
109	Ishita	NULL

4 rows in set (0.00 sec)

```
mysql> -- get all departments and the employees in them.
```

```
mysql> select e.emp_id, e.name, d.department_name from employees e right join
departments d on e.department_id = d.department_id;
```

emp_id	name	department_name
101	Arjun	Human Resources
107	Gowri	Engineering
103	Charan	Engineering
102	Bhavana	Engineering
110	Jagan	Marketing
104	Divya	Marketing
106	Farhan	Finance
108	Harish	Operations

8 rows in set (0.00 sec)

```
mysql> insert into departments values(6, 'Sales');
```

Query OK, 1 row affected (0.01 sec)

```
mysql> select e.emp_id, e.name, d.department_name from employees e right join
departments d on e.department_id = d.department_id;
```

emp_id	name	department_name
101	Arjun	Human Resources
107	Gowri	Engineering
103	Charan	Engineering
102	Bhavana	Engineering
110	Jagan	Marketing
104	Divya	Marketing
106	Farhan	Finance
108	Harish	Operations
NULL	NULL	Sales

9 rows in set (0.00 sec)

```
mysql> select e1.name as employee1, e2.name as employee2, e1.department_id from
```

```
employees e1 join e2 on e1.department_id = e2.department_id and e1.emp_id !=
e2.emp_id;
```

ERROR 1146 (42S02): Table 'cg_mysql.e2' doesn't exist

```
mysql> select e1.name as employee1, e2.name as employee2, e1.department_id from
employees e1 join employees e2 on e1.department_id = e2.department_id and e1.emp_id
!= e2.emp_id;
```

employee1	employee2	department_id
Gowri	Bhavana	2
Charan	Bhavana	2
Gowri	Charan	2
Bhavana	Charan	2
Jagan	Divya	3
Charan	Gowri	2
Bhavana	Gowri	2
Divya	Jagan	3

8 rows in set (0.00 sec)

```
mysql> select e.name, d.department_name from employees e left join departments d on
e.department_id = d.department_id
```

-> union

```
-> select e.name, d.department_name from employees e right join departments d
on e.department_id = d.department_id;
```

name	department_name
Arjun	Human Resources
Bhavana	Engineering
Charan	Engineering
Divya	Marketing
Eswar	NULL
Farhan	Finance
Gowri	Engineering
Harish	Operations
Ishita	NULL
Jagan	Marketing
NULL	Sales

11 rows in set (0.05 sec)

```
mysql> select d.department_name, count(e.emp_id) as total_employees from
departments d left join employee e on e.department_id = d.department_id group by
d.department_name;
```

ERROR 1146 (42S02): Table 'cg_mysql.employee' doesn't exist

```
mysql> select d.department_name, count(e.emp_id) as total_employees from
departments d left join employees e on e.department_id = d.department_id group by
d.department_name;
```

+-----+

department_name	total_employees
Human Resources	1
Engineering	3
Marketing	2
Finance	1
Operations	1
Sales	0

6 rows in set (0.04 sec)

```
mysql> select d.department_name, sum(e.salary) as total_salary from departments d
left join employees e on e.department_id = d.department_id group by
d.department_name;
```

department_name	total_salary
Human Resources	50000.00
Engineering	199000.00
Marketing	105000.00
Finance	55000.00
Operations	48000.00
Sales	NULL

6 rows in set (0.00 sec)

```
mysql> select e.name, count(p.project_id) as total_projects from employees e left
join projects p on e.emp_id = p.emp_id group by name;
```

name	total_projects
Arjun	0
Bhavana	2
Charan	1
Divya	1
Eswar	0
Farhan	1
Gowri	1
Harish	0
Ishita	0
Jagan	1

10 rows in set (0.00 sec)

```
mysql> select e.name, d.department, e.salary from employees e join departments d on
e.department_id = d.department_id order by e.salary;
```

ERROR 1054 (42S22): Unknown column 'd.department' in 'field list'

```
mysql> select e.name, d.department_name, e.salary from employees e join departments
d on e.department_id = d.department_id order by e.salary;
```

e.name	d.department_name	e.salary
--------	-------------------	----------

name	department_name	salary
Divya	Marketing	45000.00
Harish	Operations	48000.00
Arjun	Human Resources	50000.00
Farhan	Finance	55000.00
Jagan	Marketing	60000.00
Gowri	Engineering	62000.00
Bhavana	Engineering	65000.00
Charan	Engineering	72000.00

8 rows in set (0.00 sec)

```
mysql> select d.department_name, count(e.emp_id) as total_employees from
departments d left join employees e on e.department_id = d.department_id group by
d.department_name order by total_employees desc;
```

department_name	total_employees
Engineering	3
Marketing	2
Human Resources	1
Finance	1
Operations	1
Sales	0

6 rows in set (0.00 sec)

```
mysql> select d.department_name, count(e.emp_id) as total_employees from
departments d left join employees e on e.department_id = d.department_id group by
d.department_name where total_employees > 1 order by total_employees desc;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that
corresponds to your MySQL server version for the right syntax to use near 'where
total_employees > 1 order by total_employees desc' at line 1
mysql> select d.department_name, count(e.emp_id) as total_employees from
departments d left join employees e on e.department_id = d.department_id group by
d.department_name having total_employees > 1 order by total_employees desc;
```

department_name	total_employees
Engineering	3
Marketing	2

2 rows in set (0.00 sec)

```
mysql> select d.department_name, sum(e.salary) as total_salary from departments d
left join employees e on e.department_id = d.department_id group by
d.department_name having total_salary > 50000 order by total_salary;
```

department_name	total_salary
-----------------	--------------

Finance	55000.00
Marketing	105000.00
Engineering	199000.00

3 rows in set (0.00 sec)

mysql> notee